8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

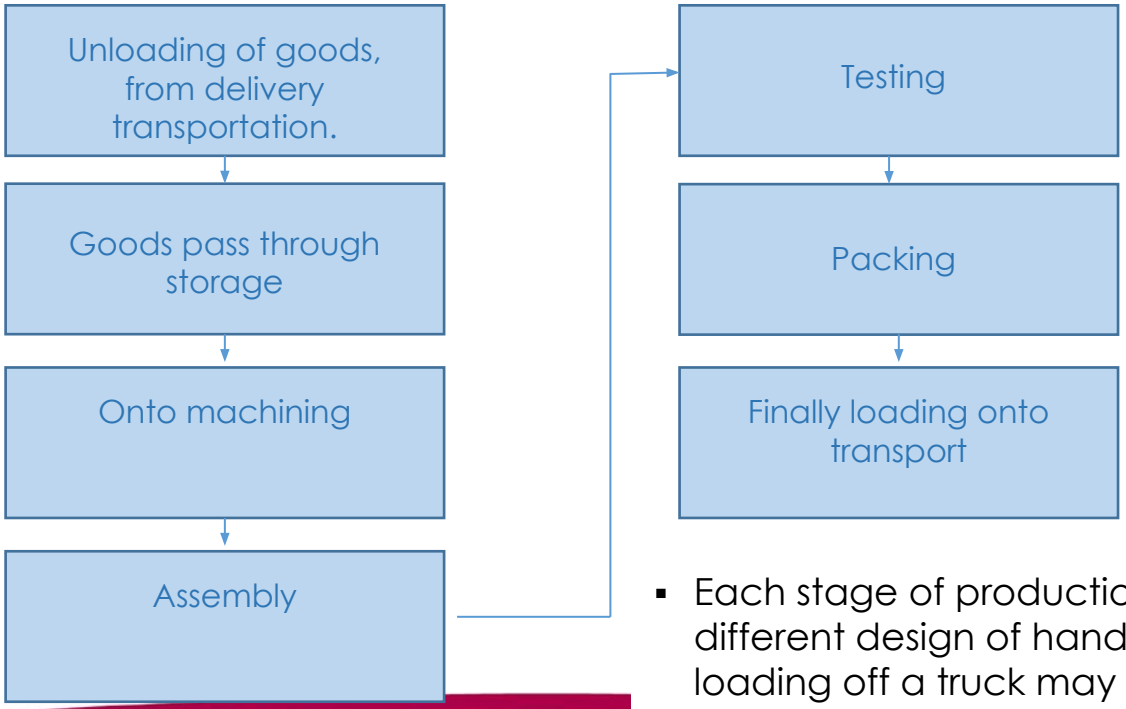
Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

c) Materials Handling Systems

Automated Storage and Retrieval Systems (ASRS), Automatic Guided Vehicles (AGVs)

Automated materials-handling systems

- Materials-handling systems provide transportation and storage of materials, components and assemblies.
- Materials handling activities start:



- Each stage of production requires a slightly different design of handling equipment, e.g. loading off a truck may require a fork lift truck.

Automated materials-handling systems

- Computer Integrated Manufacturing (CIM) systems enable a range of automated-handling systems to operate within the workplace.
- Fully automated handling systems ensure that materials, components and assemblies are delivered to the production line when required without significant manual intervention.









Automated storage and retrieval systems (ASRS)

Automated storage and retrieval systems (ASRS)

- ***Automated storage and retrieval system (ASRS)*** facilitate the storage, sorting and retrieval of components, materials and products of various sizes in computerised warehouses.



- Narrow aisles are a feature of many of these warehouses within which a variety of computer operated systems move about to store and retrieve items as required.

Advantages and Disadvantages of using Automated storage and retrieval systems (ASRS)

Advantages	Disadvantages
Small footprint – can save up to 85% of the space required	High level of customisation required, such as integration with software and hardware and the installation of conveyors.
Reduced need for labour – approximately 66% less labour required.	Expensive to install due to the complexity of the systems and the need for high level of accuracy.
Accuracy of picking items is improved – some system manufacturers claim 99.9% accuracy rate.	Highly trained operators need to be available to deal with breakdowns.
Operates faster – typically more than four times as many load movements per hour.	
Control over inventory is improved – ASRS integrates with stock control systems.	
Safety and ergonomics are improved – employees are less at risk.	

Automated storage and retrieval systems (ASRS)



Automated Guided Vehicles (AGV)

Automated Guided Vehicles (AGV)

- ***Automated Guided Vehicles (AGV)*** are computer-operated vehicles that operate within factories to move materials and parts between storage areas and work areas.
- They are often used to service flexible manufacturing cells (FMC's).
- They are guided by a variety of systems:
  - *Radio signals from wires buried in the floor*
  - *Magnetic or coloured guide tape on the floor*
  - *A laser system employing transmitters, receivers, sensors and reflectors*
  - *Gyroscopes and floor-embedded transponders*
  - *Camera based vision guidance systems*

Automated Guided Vehicles (AGV)

- Collaborative mobile robots (CMR's) are a form of AGV that has been developed to move items within workspaces, alongside people.

- They are sometimes used to follow workers to transport items and to fetch and carry in dangerous places. Some types can follow voice instructions.







Automated guided vehicles (AGV)

- AGVs are used to move parts, materials, tooling between machines or work centres.

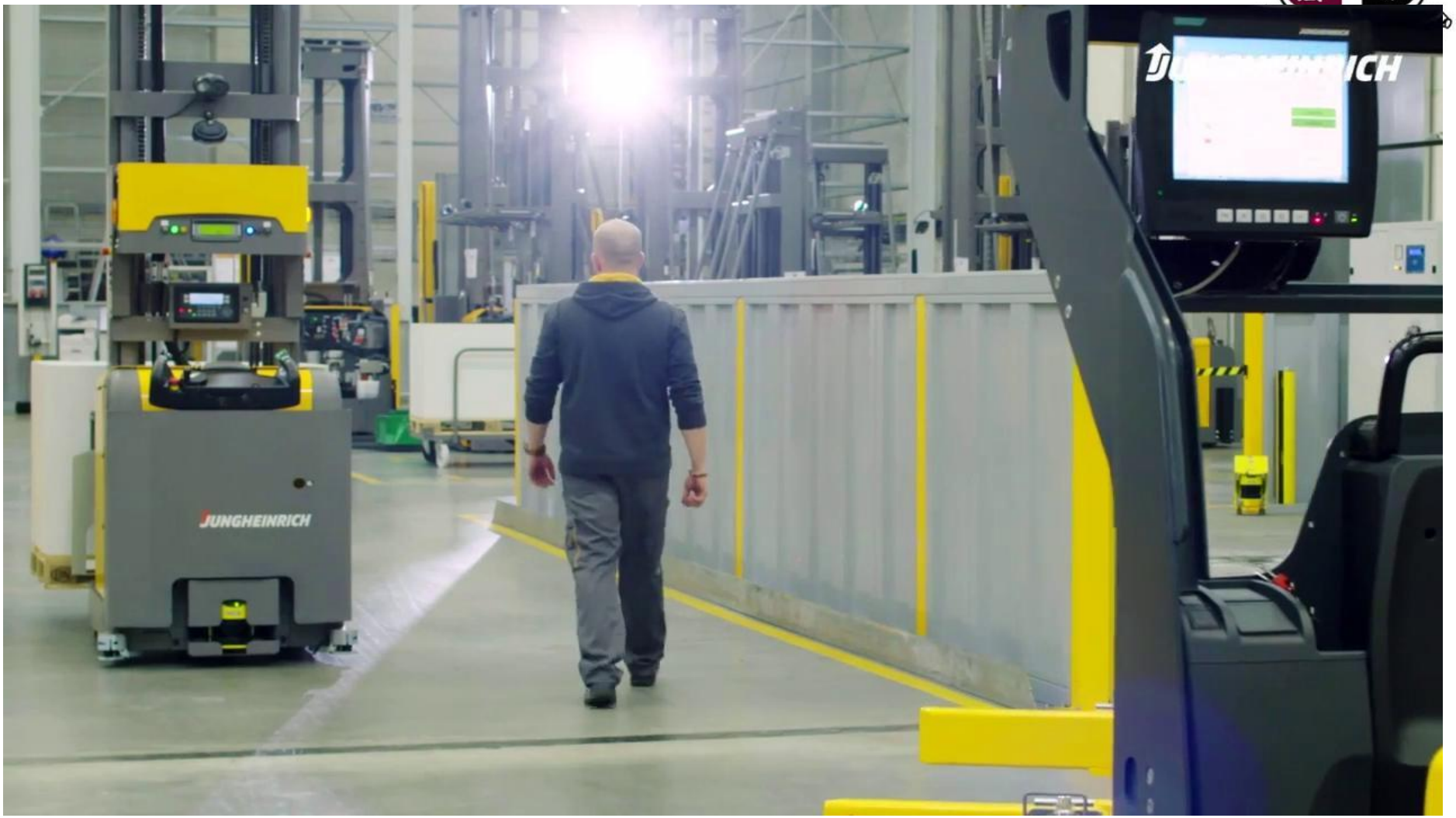
AGV Type	Application
Towing Vehicles	• These were the first type introduced and are still a very popular type today to pull a variety of trailer types.
Unit load vehicles	• These are equipped with deck that permit unit load transportation and often automatic load transfer. • Decks can be either lifted and lowered type, powered or non-powered roller, chain, belt decks or custom decks with multiple compartments.
Pallet trucks	• These are designed to transport palletised loads to and from floor level, eliminating the need for fixed load stands.
Fork truck	• These have the ability to service loads both at floor level and on stands. • In some cases these vehicles can also stack loads in a racking system.

Automated guided vehicles (AGV)

AGV Type	Application
Light load	▪ These are used to transport small parts, baskets or other light loads through a light manufacturing environment and are designed to operate in areas with limited space.
Assembly line vehicles	▪ These are an adaption of the light load AGV's for the applications involving serial assembly processes such as manufacturing cells.

- AGVs are small independently powered machines.
- Usually guided by radio frequency wires buried in the floor, optical sensors in a laser guided navigation system, GPS guided, Gyroscopic guided. Floor mounted reflective strip, tape or line guided.
- They are controlled by receiving instructions either from a central computer or from their own on board computer.

Automated guided vehicles (AGV)



Now Test Yourself:

1. What do the following terms mean and why are they both important in servicing ***flexible manufacturing cells (FMC's)***?
   - ***Automated Storage and Retrieval Systems (ASRS)***
   - ***Automated Guided Vehicles (AGV)***